SectraTEK

AGENTIC AI FOR DIGITAL FORENSICS · LAB 05

Many Agents, One Verdict



The Multiagent Pattern: specialist agents each do one part of the review, hand their findings forward, and cross-check toward a single conclusion.

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YOUR INSTRUCTOR

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IN ONE LINE

I study how phones get spied on and tampered with, and build AI tools that help examiners find the evidence.

Mobile threat research
Spyware, stalkerware and device tampering.

AI for investigators
Machine learning that speeds up the search.

Teaching
AI for Cybersecurity and Digital Forensics.

TODAY'S STORY

A team that checks itself

1

What a team adds

Specialist roles that challenge one another.

2

Before AI

Forensic labs already separate duties.

3

The problem

One agent doing everything checks nothing.

4

The discipline

Role boundaries, and each role tests the last.

5

The gap

A lone reviewer cannot see its own blind spot.

6

Multiagent

Split the work, pass context, reconcile.

7

The case

A patient file, an upload, a custody gap.

8

Your lab

Run three agents, then let a Crew orchestrate.

Practical runs in Google Colab with a free hosted model. Nothing is installed on your computer.

WHERE WE ARE

Four labs, one agent. Now, a team.

Labs 1 to 4 · One agent

A single agent reflected, used tools, reasoned step by step, and planned. Powerful, but it graded its own work.

There was no second set of eyes.

TODAY

Lab 05 · Multiagent

Now several specialist agents each take one part of the review, hand their findings forward, and challenge each other.

The team cross-checks before anything is concluded.

THE KEY IDEA

Narrow roles that hand work forward



Specialize

Each agent has one job and one set of tools, not the whole case.



Hand off

An agent's output becomes the next agent's context.



Challenge

Later roles test and narrow the earlier claims, not just agree.

EACH AGENT IS ITSELF A REACT AGENT

The team is built from Lab 03: every member reasons and calls evidence tools inside its own turn. What is new is the handoff and the challenge between members.

THE LAB ALREADY WORKS THIS WAY

Separation of duties

The examiner

Reconstructs what the device records show.

The reviewer

Independently checks the examiner's claims.

The custodian

Tracks who handled the evidence, and when.

The sign-off

One conclusion, only after all three agree.

NO SINGLE PERSON OWNS THE WHOLE CHAIN

A real forensic lab splits examination, review, and evidence custody across people on purpose, so one person's blind spot does not become the finding. Multiagent teaches a model team the same separation of duties.

THE PROBLEM

One agent doing everything checks nothing

ONE AGENT, EVERY HAT

"I found an upload and the custody log looks fine, so the patient file was transmitted."

WHY A SOLO AGENT FAILS HERE

It grades its own work

The claim and the check come from the same reasoning.

It blends two questions

Transmission evidence and custody handling get mixed together.

It misses the boundary

"upload started" is quietly treated as "upload completed."

THE DISCIPLINE

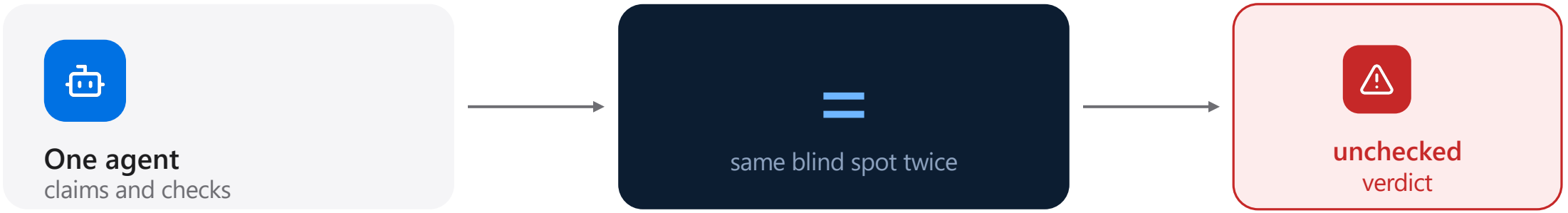
Give each role a boundary, and a critic

1	**One job per agent**	An investigator builds; it does not also sign off on itself.
2	**Independent verification**	A second role tests the claim against the raw records.
3	**Keep questions apart**	The technical question and the custody question stay separate.
4	**Reconcile, then conclude**	One final answer, written only after the roles are compared.

The value is not more agents. It is the independent check between them.

THE GAP

A lone reviewer cannot audit itself

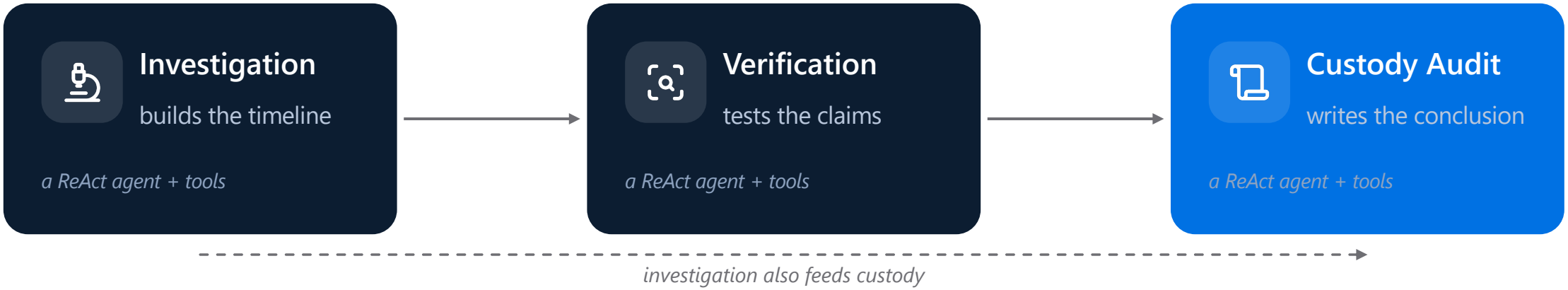


INDEPENDENCE IS THE WHOLE VALUE

If the same reasoning makes a claim and checks it, a blind spot survives both. Only a genuinely separate role, with its own task and its own look at the records, can catch an overreach the first agent could not see. That is why the team passes work between distinct agents, not back to the same one.

THE ANSWER

The Multiagent Pattern



CONTEXT FLOWS FORWARD

Each agent's output becomes the next agent's context. Custody sees both earlier findings.

A CREW RUNS THE ORDER

The Crew sorts the agents by dependency and runs each one once its inputs are ready.

WHAT MOVES BETWEEN THEM

The handoff is the point



Investigation hands forward

A draft timeline and a first transmission assessment.



Verification hands forward

A corrected rating: upload started, not completed.



Custody adds the last layer

The handling record, and how its gap affects confidence.

NOTHING IS LOST BETWEEN ROLES

Because each output becomes the next role's context, the final conclusion carries the whole chain of reasoning, built by three independent specialists.

THE INVESTIGATION

Case MA-2026-022

THE DEVICE

PHONE

Motorola Edge 2024, Android 14

SETTING

outreach phone, vaccination event

THE FILE

patients_contacts.png

THE CONCERN

was it transmitted?

WHAT THE RECORDS SHOW

A file

patients_contacts.png created at 18:44, inside the window.

An upload started

SecureChat attach at 18:45, then upload_started, 48 KB out.

A custody gap

The handling log is missing one analyst-to-analyst transfer.

KEEP THEM APART

Two questions, not one

A TECHNICAL question

Was the file transmitted?

- File created, attach attempt, upload_started.
- No upload_completed record exists.
- Strongest honest label: LIKELY, not confirmed.

A CUSTODY question

Was the evidence handled well?

- One analyst-to-analyst transfer is undocumented.
- That weakens confidence in the handling.
- It is NOT proof about what the phone did.

A custody gap lowers confidence in the evidence. It does not turn a likely transmission into a confirmed one.

YOUR TEAM

Three agents, two tool kits

InvestigationAgent

technical kit: device, file, messaging, network

Builds the timeline. Should treat upload_started as an attempt, not completion.

EvidenceVerificationAgent

the same technical kit

Re-opens the records. Should lower the rating to LIKELY when it sees no completion.

CustodyAuditAgent

custody kit: file list and custody log

Audits handling. Writes the final label and explains the custody gap.

The custody agent gets only the custody kit. Role boundaries are enforced by the tools each agent holds.

PRACTICAL

Your turn, in the browser

1	Open + Part 0 setup	Click the Colab badge. Steps 0.1 to 0.4: install, paste a free Groq key (hidden, not saved), write the files, load the team code.
2	Part A: one at a time	A.1 makes two tool kits, A.2 builds and wires the three agents. A.3 to A.5 run them in order: watch the context pass forward.
3	Part B: the Crew	B.1 builds a fresh team inside a Crew and prints the run order. B.2 runs the whole team in one go.
4	Your Final Judgment	Four points: the transmission label and why, the records behind it, the custody impact kept separate, what stays unproven.
5	What to notice	Four checks: no finished upload claimed, confidence lowered, custody gap not used as proof, the two questions kept apart.

Before every code cell, a short note tells you its Purpose, its Input, How it works, and its Output.

SectraTEK

Five patterns, one habit: stay inside the evidence.

You have built Reflection, Tool Use, ReAct, Planning, and Multiagent. Each makes an AI a better assistant to an examiner, never the examiner.

